



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

15-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	C	C	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C,D	A,B,C	A,D	A,B	A,B,C	A,B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	20.00	0.30	0.00	2.00	2.00	9.00
SECTION-II (ii)	Q.	7	8	9			
	A.	2	12	8			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	D	C	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C	A,B,D	B,C,D	A,D	A,C,D	A,B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	5.18	10.00	2.00	3.00	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	4	4	2			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	C	C	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B	A,C	B,D	A,B,C,D	A,B,C	A,B
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	2.00	0.50	-0.50	0.12 or 0.13	0.33
SECTION-II (ii)	Q.	7	8	9			
	A.	1	4	9			

HINT – SHEET

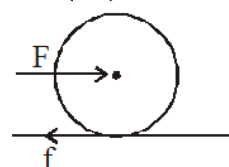
PART-1 : PHYSICS

SECTION-I (i)

2. Ans (C)

$$I = \frac{I_0}{2} \cos^2 \theta \sin^2 \theta = \frac{I_0}{8} \sin^2 (2\theta)$$

4. Ans (D)



$$F - f = Ma \quad \dots\dots (1)$$

$$fR = \frac{MR^2\alpha}{2} \quad \dots\dots (2)$$

$$a = R\alpha \quad \dots\dots (3)$$

From 1st

$$F - f = 2f$$

$$F = 3f$$

$$f_{\max} = \mu_s mg \text{ so } F = 3\mu_s mg$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,B,C,D)

$$\tan i_p = \mu = \sqrt{3}$$

$$i_p = 60^\circ$$

$$1 \sin 60 = \sqrt{3} \sin \theta$$

$$\theta = 30^\circ$$

$$\tan i_p' = \frac{1}{\sqrt{3}}$$

$$i_p' = 30^\circ$$

$\Rightarrow$ 4 is completely polarized

6. Ans (A,B,C)

$$\frac{\partial E}{\partial x} = -\frac{\partial B}{\partial t}$$

$$\frac{\partial B}{\partial x} = -\frac{1}{C^2} \frac{\partial E}{\partial t}$$

$$-\frac{1}{9 \times 10^{16}} \times 100 \times 10^9 = \frac{\partial B}{\partial x}$$

$$\& \frac{E}{B} = C$$

7. Ans (A,D)

$$\vec{B} = \frac{E}{C} \hat{k} \sin(\pi x - 3\pi \times 10^8 t) = 10^{-5} T \hat{k} (\sin(\pi x - \omega t))$$

$$\phi = \int B dA = 2 \int_1^2 B dx$$

$$= 2 \times 10^{-5} \left[\int_1^2 \sin(\pi x - \omega t) dx \right]$$

$$= 2 \times 10^{-5} \left[-\cos \frac{(\pi x - \omega t)}{\pi} \right]_1^2$$

$$= \frac{-2 \times 10^{-5}}{\pi} [2 \cos \omega t] = -\frac{4}{\pi} \times 10^{-5} \cos \omega t$$

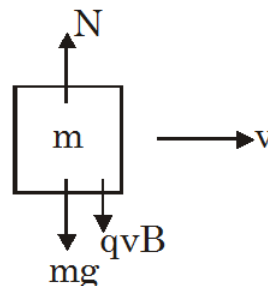
$$r = \frac{d\phi}{dt} = -\frac{4}{\pi} \times 10^{-5} \times 3\pi \times 10^8 \sin \omega t = -12000 \sin \omega t$$

9. Ans (A,B,C)

$$dF = i (\vec{dl} \times \vec{B})$$

here $dl \parallel B$, hence $F = 0$

10. Ans (A,B,C)



Using F.B.D.

Magnitude of velocity is maximum at mean position. Hence normal force will be maximum and minimum at mean position.

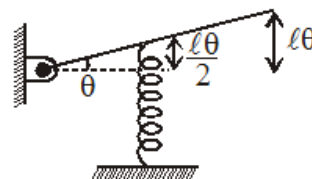
Also at extreme position, speed will be zero

$$\therefore N = mg$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (20.00)



$$x = \frac{\ell}{2} \theta$$

$$v = \frac{d}{dt}(\ell \theta) = \ell \Omega$$

$$I \frac{d^2 \theta}{dt^2} = -k \frac{\ell}{2} \theta \times \frac{\ell}{2} - c \times v \times \ell$$

$$\frac{m \ell^2}{3} \frac{d^2 \theta}{dt^2} + c \ell^2 \frac{d\theta}{dt} + \frac{k \ell^2}{4} \theta = 0$$

$$\frac{d^2 \theta}{dt^2} + \frac{3c}{m} \frac{d\theta}{dt} + \frac{3k}{4m} \theta = 0$$

$$\omega = \sqrt{\frac{3k}{4m} - \left(\frac{3c}{2m}\right)^2} = \sqrt{\frac{3k}{4m} - \frac{9}{4} \frac{c^2}{m^2}}$$

$$= \sqrt{\frac{3}{4} \times \frac{25}{3} \frac{c^2}{m^2} - \frac{9}{4} \frac{c^2}{m^2}}$$

$$= \frac{2c}{m} = 20 \text{ rad/s}$$

2. **Ans (0.30)**

$$\frac{m\ell^2}{3} \frac{d^2\theta}{dt^2} + c\ell^2 \frac{d\theta}{dt} + kx^2\theta = 0$$

if critically damped, $\omega_0 = r$

$$\sqrt{\frac{3kx^2}{m\ell^2}} = \frac{3c\ell^2}{2m\ell^2}$$

$$\Rightarrow \sqrt{3 \times \frac{25}{3} \frac{c^2}{m^2} \frac{x^2}{\ell^2}} = \frac{3c}{2m}$$

$$x = \frac{3}{10} \ell = 0.3\ell$$

3. **Ans (0.00)**

$$Q = 2\pi \times \frac{\frac{1}{2}m\omega^2 A^2}{P \times T} = \frac{\omega m}{b}$$

$$\Rightarrow \frac{\omega^2 A^2 \times 6\pi\epsilon_0 C^3}{2 \times q^2 \times \omega^4 A^2} = \frac{1}{b}$$

4. **Ans (2.00)**

$$Q = 2\pi \times \frac{\frac{1}{2}m\omega^2 A^2}{P \times T} = \frac{\omega m}{b}$$

$$\Rightarrow \frac{\omega^2 A^2 \times 6\pi\epsilon_0 C^3}{2 \times q^2 \times \omega^4 A^2} = \frac{1}{b}$$

5. **Ans (2.00)**

$$(V_L - V_C) = 50$$

$$V_R^2 + (V_L - V_C)^2 = 130^2$$

$$\Rightarrow V_R^2 = 130^2 - 50^2$$

$$\therefore \text{Power factor} = \frac{V_R}{V} = \frac{120}{130} = \frac{12}{13}$$

6. **Ans (9.00)**

$$V_R^2 + V_L^2 = 150^2$$

$$\Rightarrow V_L = \sqrt{150^2 - 120^2} = 90V$$

$$V_L - V_C = 50$$

$$\Rightarrow V_C = 90 - 50 = 40V$$

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (2)**

$a_{\max}$ is at extreme.

$$a = a_0 e^{-\tau t}$$

$$a = a_0 e^{-n\tau t}$$

$$a = a_0 e^{-\ell n s^n}$$

$$\frac{a_0}{64} = \frac{a_0}{8^n}$$

$$\Rightarrow n = 2$$

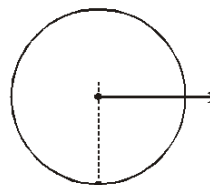
8. **Ans (12)**

$$P = \frac{\left(\frac{d^2 p}{dt^2}\right)^2}{6\pi\epsilon_0 C^3} = \frac{p_0^2 \omega^4 \sin^2 \omega t}{6\pi\epsilon_0 C^3}$$

$$\text{average of } \sin^2 \omega t = \frac{1}{2}$$

$$\Rightarrow P_{\text{av}} = \frac{p_0^2 \omega^2}{12\pi\epsilon_0 C^3}$$

9. **Ans (8)**



$$Cm2mv_C = mv$$

$$\Rightarrow v_C = \frac{v}{2}$$

about new Cm

$$L_i = mv \frac{a}{2} = \left(\frac{ma^2}{2} + m \left(\frac{a}{2} \right)^2 \right) \omega + m \left(\frac{a}{2} \right)^2 \omega$$

$$\omega = \frac{v}{2a}$$

$$k_f = \frac{1}{2} 2m \times \left(\frac{v}{2} \right)^2 + \frac{1}{2} \times ma^2 \times \left(\frac{v}{2a} \right)^2 = \frac{3}{8} mv^2$$

$$\Delta E = \frac{mv^2}{8}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (C)**

$$\ell n K = \ell n A - \frac{E_a}{RT}$$

$$\text{slope} = \frac{-E_a}{R}$$

$\ell n K (s^{-1})$	$\frac{1}{T} (in K^{-1})$	$T (in K)$
4	0.004	250K
5	0.003	333.33K
6	0.002	500K
7	0.001	1000K

$$\text{Slope} = \frac{-E_a}{R} = \frac{-1}{0.001}$$

$$\Rightarrow E_a = R \times 1000$$

$$= 2 \text{ Kcal/mole}$$

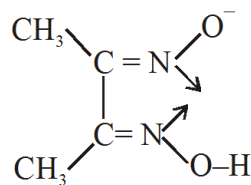
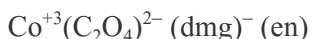
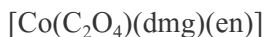
PART-2 : CHEMISTRY

SECTION-II (i)

1. **Ans (2.00)**

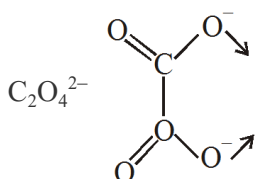
cd

4. Ans (2.00)

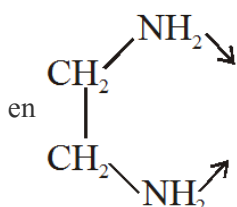


Unsymmetrical ligand

BC
dmg[⊖]



Symmetrical ligand AA

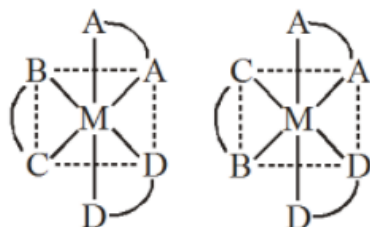


Symmetrical ligand DD

Co-ordination compounds becomes



All three are bidentate ligands



2 Geometrical isomers are possible.

PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (4)

$$E_{\text{a net}} = \frac{K_1 E_{\text{a}_1} + K_2 E_{\text{a}_2} + K_3 E_{\text{a}_3}}{K_1 + K_2 + K_3}$$

$$K_1 : K_2 : K_3 = 2/3 : 3/2 : 4$$

Using these values =

$$\frac{4 \times 8 + 9 \times 6 + 24 \times 3}{4 + 9 + 24} = 4.27 \text{ Kcal}$$

9. Ans (2)

For ppts of MS

$$[\text{M}^{2+}] [\text{S}^{--}] = 10^{-19}$$

$$[\text{S}^{--}] = 10^{-18}$$

$$K_{\text{a}_1} \cdot K_{\text{a}_2} = \frac{[\text{H}^+]^2 [\text{S}^{--}]}{[\text{H}_2\text{O}]}$$

$$10^{-21} = \frac{[\text{H}^+]^2 \times 10^{-18}}{0.1}$$

$$[\text{H}^+] = 10^{-2} \text{M}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)

$$\therefore 2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$\Rightarrow 2.4 \frac{dx}{dt} + 2.3 (-1) = 0 \Rightarrow \frac{dx}{dt} = \frac{3}{4}$$

2. Ans (C)

$$\lim_{x \rightarrow -\infty} \sqrt{x^2 - \frac{4x}{3}} + x + 2$$

$$\lim_{x \rightarrow -\infty} \left(\frac{x^2 - \frac{4x}{3} - (x+2)^2}{\sqrt{x^2 - \frac{4x}{3}} - (x+2)} \right)$$

$$\lim_{x \rightarrow -\infty} \frac{-\frac{16}{3}x - 4}{|x| \sqrt{1 - \frac{4}{3x}} - (x+2)}$$

$$\lim_{x \rightarrow -\infty} \frac{x \left(-\frac{16}{3} - \frac{4}{x} \right)}{x \left[-\sqrt{1 - \frac{4}{3x}} - \left(1 + \frac{2}{x} \right) \right]}$$

$$\lim_{x \rightarrow -\infty} -\frac{16}{3(-2)} = \frac{8}{3}$$

3. Ans (C)

$$f(x) + f(y) = f(x)f(y) + f(xy)$$

partial differentiation wrt x

$$f'(x) = f'(x)f(y) + f'(xy).(y)$$

put x = 1

$$-2 = -2f(y) + f'(y)y$$

$$\text{get } f(y) = 1 - y^2$$

$$\Rightarrow f(x) = 1 - x^2$$

4. Ans (C)

$$x = 1, f(1) + 2f(1) = 1$$

$$\Rightarrow f(1) = \frac{1}{3}$$

$$f(x^2) + 2x^2 f'(x^2) + (x^2 + 1)f'(x) + 2xf'(x) = 1$$

$$x = 1$$

$$f(1) + 2f'(1) + 2f'(1) + 2f(1) = 1$$

$$4f'(1) + 1 = 1 \Rightarrow f'(1) = 0$$

$$2xf'(x^2) + 4x^3 f''(x^2) + 4xf'(x^2) + 2xf'(x)$$

$$+ (x^2 + 1)f''(x) + 2f(x) + 2f'(x) = 0$$

$$x = 1$$

$$2f'(1) + 4f''(1) + 4f'(1) + 2f'(1) + 2f''(1)$$

$$+ 2f(1) + 2f'(1) = 0$$

$$6f''(1) + \frac{2}{3} = 0$$

$$f''(1) = -\frac{1}{9}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B)

$$14 = 2^x + 4^x + 8^x$$

$$\Rightarrow x = 1$$

$$f'(x) = 2^x \ln 2 + 4^x \ln 4 + 8^x \ln 8$$

$$f'(1) = 2 \ln 2 + 8 \ln 2 + 24 \ln 2$$

$$f'(1) = 34 \ln 2$$

$$g'(14) = \frac{1}{34(\ln 2)}$$

6. Ans (A,C)

$$\because \sqrt{y^2 + 2x} + \sqrt{y^2 - 2x} = 2x^2$$

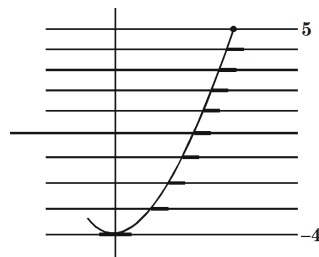
$$\because \sqrt{y^2 + 2x} - \sqrt{y^2 - 2x} = \frac{2}{x}$$

$$\Rightarrow y^2 + 2x = \left(x^2 + \frac{1}{x}\right)^2$$

$$\text{or } y^2 = x^4 + \frac{1}{x^2}$$

$$\Rightarrow y = \sqrt{x^4 + \frac{1}{x^2}}$$

7. Ans (B,D)



$$f(x) = [x^2 - 4] \forall x \in \left[-\frac{1}{3}, 3\right]$$

$$\Rightarrow x^2 - 4 \in [-4, 5]$$

discontinuous at 9 points

$$g(x) = [x^2 - 4](|x - 2| + |3x - 5|)$$

8. Ans (A,B,C,D)

$$\sin(\tan^{-1} x) = \frac{x}{\sqrt{1+x^2}},$$

$$\sin(\cos^{-1} x) = \cos(\sin^{-1} x) = \sqrt{1-x^2}$$

Hence, equation reduces to

$$\frac{x}{\sqrt{1+x^2}} = \sqrt{4 - (2\sqrt{1-x^2})^2} = 2|x|$$

$$\frac{x}{\sqrt{1+x^2}} = 2|x| \Rightarrow x = 0 \text{ is the only solution}$$

$$A) \sin^{-1} a + \cos^{-1} a = \frac{\pi}{2} \quad (\text{always true})$$

$$B) 2\sin^{-1} a + \cos^{-1} a = \frac{\pi}{2} + \sin^{-1} a = \frac{\pi}{2} \quad (\text{True})$$

$$C) \sin^{-1} a + 3\cos^{-1} a = \frac{\pi}{2} + 2\cos^{-1} a = \frac{3\pi}{2} \quad (\text{True})$$

$$D) \tan^{-1} a + \cos^{-1} a = \frac{\pi}{2} \quad (\text{True})$$

9. Ans (A,B,C)

$$-1 \leq \sum_{k=1}^n 3^k (f(x+ky) - f(x-ky)) \leq 1$$

Also,

$$-1 \leq \sum_{k=1}^{n-1} 3^k (f(x+ky) - f(x-ky)) \leq 1$$

$$-2 \leq 3^n (f(x+ny) - f(x-ny)) \leq 2$$

$$\frac{-2}{3^n} \leq (f(x+ny) - f(x-ny)) \leq \frac{2}{3^n}$$

$$\Rightarrow f(x+ny) - f(x-ny) = 0$$

$$f'(x+ny) + f'(x-ny) = 0$$

$$y \rightarrow 0 \Rightarrow f'(x) = 0$$

$$\Rightarrow f(x) \text{ is constant function}$$

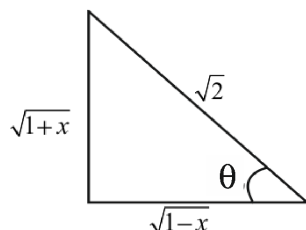
10. Ans (A,B)

$$\sin^{-1} \sqrt{\frac{1+x}{2}} - \sqrt{2-x} = \cot^{-1}(\tan \sqrt{2-x}) - \sin^{-1} \sqrt{\frac{1-x}{2}}$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$-1 \leq x \leq 1 \quad x \leq 2 \quad x \leq 2 \quad -1 \leq x \leq 1$$

Domain : $x \in [-1, 1]$



$$\sin^{-1} \sqrt{\frac{1+x}{2}} + \sin^{-1} \sqrt{\frac{1-x}{2}} - \cot^{-1}(\tan \sqrt{2-x}) = \sqrt{2-x}$$

$$\frac{\pi}{2} - \cot^{-1}(\tan \sqrt{2-x}) = \sqrt{2-x}$$

$$\tan^{-1}(\tan \sqrt{2-x}) = \sqrt{2-x}$$

Let $\sqrt{2-x} = \alpha$ where $\alpha \in [1, \sqrt{3}]$

$$\Rightarrow \tan^{-1}(\tan \alpha) = \alpha \quad \dots\dots(1)$$

Case I : $\alpha \in [1, \frac{\pi}{2})$

$$\tan^{-1}(\tan \alpha) = \alpha \text{ happens } \forall \alpha \in [1, \frac{\pi}{2})$$

$$1 \leq \sqrt{2-x} < \frac{\pi}{2} \text{ or } 2 - \frac{\pi^2}{4} < x \leq 1$$

Case II: $\alpha \in [\frac{\pi}{2}, \sqrt{3}]$

$$\tan^{-1}(\tan \alpha) \text{ becomes } \alpha - \pi$$

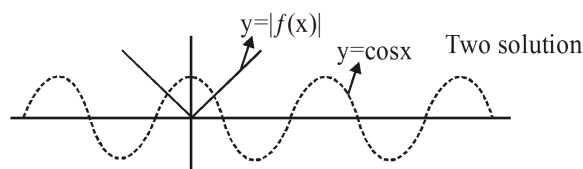
Hence (1) becomes $\alpha - \pi = \alpha$ (Not possible)

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (2.00)

$$f(x) = 2x$$



2. Ans (2.00)

$h(x) = |(2x)(e^{2x} - 1)(2x - 1)(2x - 2)|$ is not differentiable at $x = \frac{1}{2}$ & $x = 1$.

3. Ans (0.50)

$$y = \cot^{-1} \left(\frac{2 + 2\sqrt{\cos^2 x}}{2 \sin x} \right)$$

$$\Rightarrow y = \cot^{-1} \left(\frac{1 + |\cos x|}{\sin x} \right)$$

For $x \in (0, \frac{\pi}{2})$,

$$y = \cot^{-1} \left(\frac{1 + \cos x}{\sin x} \right) = \cot^{-1} \left(\cot \frac{x}{2} \right)$$

$$\Rightarrow y = \frac{x}{2} \quad \therefore \frac{dy}{dx} = \frac{1}{2}$$

$$\text{For, } x \in \left(\frac{\pi}{2}, \pi \right) \quad y = \cot^{-1} \left(\frac{1 - \cos x}{\sin x} \right)$$

$$\Rightarrow y = \cot^{-1} \left(\tan \frac{x}{2} \right)$$

$$\Rightarrow y = \cot^{-1} \left(\cot \left(\frac{\pi}{2} - \frac{x}{2} \right) \right)$$

$$\Rightarrow y = \frac{\pi}{2} - \frac{x}{2}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{2}$$

4. Ans (-0.50)

$$y = \cot^{-1} \left(\frac{2 + 2\sqrt{\cos^2 x}}{2 \sin x} \right)$$

$$\Rightarrow y = \cot^{-1} \left(\frac{1 + |\cos x|}{\sin x} \right)$$

For $x \in (0, \frac{\pi}{2})$,

$$y = \cot^{-1} \left(\frac{1 + \cos x}{\sin x} \right) = \cot^{-1} \left(\cot \frac{x}{2} \right)$$

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$$\text{For, } x \in \left(\frac{\pi}{2}, \pi \right) \quad y = \cot^{-1} \left(\frac{1 - \cos x}{\sin x} \right)$$

$$\Rightarrow y = \cot^{-1} \left(\tan \frac{x}{2} \right)$$

$$\Rightarrow y = \cot^{-1} \left(\cot \left(\frac{\pi}{2} - \frac{x}{2} \right) \right)$$

$$\Rightarrow y = \frac{\pi}{2} - \frac{x}{2}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{2}$$

5. Ans (0.12 or 0.13)

At $x = 1, y = -2$

$$\left. \frac{dy}{dx} \right|_{x=1} = \frac{2}{3} \Rightarrow f'(1) = \frac{2}{3} \text{ and } f(1) = -2.$$

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{(e^{\sin \pi x} - 1) \cdot \sin \frac{\pi}{x}}{\tan^{-1}(x-1) \cdot (f^2(x^3) - 4)} \\ = \lim_{x \rightarrow 1} \left(\frac{e^{\sin \pi x} - 1}{\sin \pi x} \right) \cdot \frac{\sin \pi x}{x-1} \cdot \frac{\sin \frac{\pi}{x}}{f^2(x^3) - 4} \\ = -\pi \lim_{x \rightarrow 1} \frac{\sin \frac{\pi}{x}}{f^2(x^3) - 4} \cdot \left(\text{form } \frac{0}{0} \right) \end{aligned}$$

using L'Hôpital

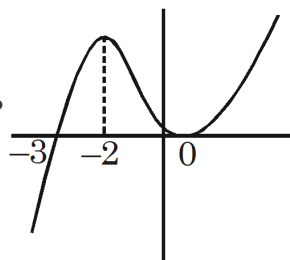
$$= -\pi \lim_{x \rightarrow 1} \frac{\cos \frac{\pi}{x} \left(-\frac{\pi}{x^2} \right)}{2f(x^3) \cdot f'(x^3) \cdot 3x^2} = \frac{\pi^2}{8}$$

6. Ans (0.33)

Any point on the curve $P(t, t^3 + 3t^2)$

$$\therefore \left. \frac{dy}{dx} \right|_P = 3t^2 + 6t$$

$\therefore$ equation of tangent at P



$$y - (t^3 + 3t^2) = (3t^2 + 6t)(x - t)$$

If this tangent passes through $(\lambda, 0)$, then

$$-t^2(t+3) = 3t(t+2)(\lambda-t)$$

$$\Rightarrow t = 0 \text{ and } 2t^2 + 3(1-\lambda)t - 6\lambda = 0 \begin{cases} t_1 \\ t_2 \end{cases}$$

$$\therefore t_1 + t_2 = \frac{3(\lambda-1)}{2} \text{ and } t_1 t_2 = -3\lambda$$

also tangents are perpendicular

$$\Rightarrow (3t_1^2 + 6t_1) \cdot (3t_2^2 + 6t_2) = -1$$

$$\Rightarrow 9t_1 t_2 (t_1 + 2)(t_2 + 2) = -1$$

$$\Rightarrow 9t_1 t_2 (t_1 + t_2 + 2(t_1 + t_2) + 4) = -1$$

$$\Rightarrow -9.3\lambda(3\lambda - 3\lambda - 3 + 4) = -1 \Rightarrow \lambda = \frac{1}{27}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (1)

$$\ell_1 = (x \cos x)^x \quad (0^0)$$

$$\begin{aligned} \log \ell_1 &= \lim_{x \rightarrow 0^+} \frac{\ln x \cos x}{1/x} \\ &= \frac{1}{x \cos x} (\cos x - x \sin x) \\ &= \frac{-1/x^2}{\cos x} \\ &= \lim_{x \rightarrow 0} \frac{-x (\cos x - x \sin x)}{\cos x} = 0 \end{aligned}$$

$$\ell_1 = e^0 = 1$$

$$\Rightarrow \ell_2 = \lim_{x \rightarrow 0^+} (x \sin x)^{1/x} = 0^{+\infty} = 0$$

$$\ell_1 + \ell_2 = 1$$

8. Ans (4)

$$x = 3 \cos \theta - \cos^3 \theta \Rightarrow \frac{dx}{d\theta} = -3 \sin^3 \theta$$

$$y = 3 \sin \theta - \sin^3 \theta \Rightarrow \frac{dy}{d\theta} = 3 \cos^3 \theta$$

$$\therefore \frac{dy}{dx} = -\frac{\cos^3 \theta}{\sin^3 \theta}$$

equation of normal is

$$y - (3 \sin \alpha - \sin^3 \alpha) = \frac{\sin^3 \alpha}{\cos^3 \alpha} (x - (3 \cos \alpha - \cos^3 \alpha))$$

$$\Rightarrow y \cos^3 \alpha - x \sin^3 \alpha = \frac{3}{4} \sin 4\alpha$$

$$\Rightarrow 4 \left[\frac{\cos^4 \alpha}{\cos \alpha} \cdot y - \frac{\sin^4 \alpha}{\sin \alpha} \cdot x \right] = 3 \sin 4\alpha$$

$$\therefore a = \cos^4 \alpha, b = \sin^4 \alpha, a + b = \sin 4\alpha$$

$$\Rightarrow \cos^4 \alpha + \sin^4 \alpha = \sin^4 \alpha$$

$$\Rightarrow 2 - \sin^2 2\alpha = 2 \sin 4\alpha$$

$$\Rightarrow 4 - [1 - \cos 4\alpha] = 4 \sin 4\alpha$$

$$\Rightarrow 3 = -\cos 4\alpha + 4 \sin 4\alpha$$

$$\Rightarrow \text{Total 4 values of } \alpha$$

9. Ans (9)

$f(x)$ is discontinuous at all integers except at $x = 0$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

15-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,D	B,C	C,D	A,B,D	A,B	C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	A	C	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	7.00 to 7.30	9.81 to 9.82	0.15	18.00	-10.00	4.40 to 4.50
SECTION-II (ii)	Q.	7	8	9			
	A.	100	3600	640			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C	A,B,D	B,C,D	A,C,D	A,B	A,B,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	A	B	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.30	8.00	3.00	8.00	2.00	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	13	36	2			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	A,C,D	A,C,D	A,B,C,D	A,C	B
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	C	B	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.50	0.93 or 0.94	2.00	3.00	6.00	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	5	1	32			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

3. Ans (C,D)

$$B = \frac{\mu_0}{4\pi X^3} m \sqrt{1 + 3\cos^2\theta}$$

$$B_{\text{axis}} = \frac{\mu_0 \times 1 \times (\pi \times (10^{-2})^2) \times 2}{4\pi \times 1^3} = 2\pi \times 10^{-11}$$

4. Ans (A,B,D)

$$i_{\text{rms}} = 5A$$

$$\text{Reading of voltmeter} = i_{\text{rms}} \sqrt{R^2 + X_C^2}$$

$$\text{From this } X_C = 20\Omega \Rightarrow \omega = 10^3 \text{ rad/s}$$

$$\Rightarrow f = \frac{1000}{2\pi}$$

$$\therefore X_L = \omega L = 20\Omega$$

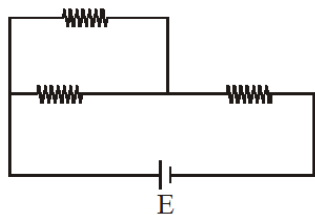
therefore the circuit is in resonance

$$E_{\text{rms}} = i_{\text{rms}} R = 50 \text{ V also power factor} = 1$$

$$\text{av. Power} = E_{\text{rms}} i_{\text{rms}} \cos \phi = 250 \text{ W}$$

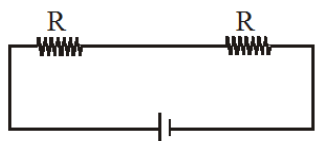
6. Ans (C)

$$t = \infty$$



$$i = \frac{E}{\frac{3R}{2}} = \frac{2E}{3R}; v_{YZ} = R \times \frac{2E}{3R} = \frac{2E}{3}$$

$$t = 0$$

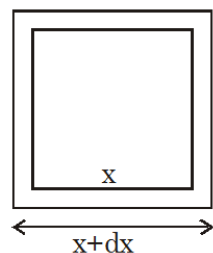


$$i = \frac{E}{2R}; v_{xy} = \frac{E}{2}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)



$$\varepsilon = x^2 \frac{dB}{dt}$$

$$x^2 \omega B_0 \cos \omega t$$

$$dP = \frac{\varepsilon^2}{dR} = \frac{\varepsilon^2}{\frac{P \times 4x}{b \frac{dx}{2}}} = \frac{b \varepsilon^2 dx}{8P x}$$

$$P = \frac{b}{8P} \omega^2 B_0^2 \cos^2 \omega t \int_0^\ell x^3 dx$$

$$= \frac{b \omega^2 B_0^2 \cos^2 \omega t \ell^4}{32P}$$

$$P_{av} = \frac{\sigma b \omega^2 B_0^2 \ell^4}{64}$$

8. Ans (A)

$$P \propto \ell^4$$

$$P_1 = \left(\frac{P_0}{4^4} \right)$$

$$P_T = 16 \times \frac{P_0}{4^4} = \frac{P_0}{16}$$

10. Ans (B)

$$(1) \frac{\omega}{k} = C = 3 \times 10^8 \text{ m/s}$$

$$\vec{k} = -(3\pi\hat{i} + 4\pi\hat{k}) \Rightarrow |\vec{k}| = 5\pi$$

$$\Rightarrow \omega = 15\pi \times 10^8 \text{ rad/s}$$

$$(2) \frac{|\vec{k}|}{C} \vec{E} = \vec{B} \times \vec{k} \Rightarrow \vec{E} = \frac{C}{|\vec{k}|} (\vec{B} \times \hat{k})$$

$$= \frac{3 \times 10^8}{5\pi} (10^{-8} \hat{j} \times (-3\pi\hat{i} - 4\pi\hat{k}))$$

$$= -\frac{3}{5} [3\hat{k} - 4\hat{i}] = 2.4\hat{i} - 1.8\hat{k} \text{ N/C}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (7.00 to 7.30)

$$i = \frac{V}{Z} \sin(\omega t - \phi)$$

$$\tan \phi = \frac{\omega L - \frac{1}{\omega C}}{R}$$

$$1 = \frac{10\pi \times 10^{-2} - \frac{k}{10\pi}}{1.6 \times 2 \times 10^{-2}}$$

$$q = \frac{V}{Z\omega} \cos(\omega t - \phi)$$

$$\tan \phi = \frac{\left(\omega L - \frac{1}{\omega C} \right)}{R}$$

$$Z = \sqrt{\left(\omega L - \frac{1}{\omega C} \right)^2 + R^2} = \sqrt{2}R$$

$$\frac{R}{2L} = r = 1.6$$

$$R = 2m \times 1.6$$

$$5 \times 10^{-2} = \frac{F_0}{Z\omega}$$

$$\Rightarrow F_0 = \frac{5 \times 10^{-2} \times \sqrt{2} \times 2 \times 10^{-2} \times 1.6}{10\pi}$$

$$= \frac{16\sqrt{2}}{\pi} \times 10^{-5} \text{ N}$$

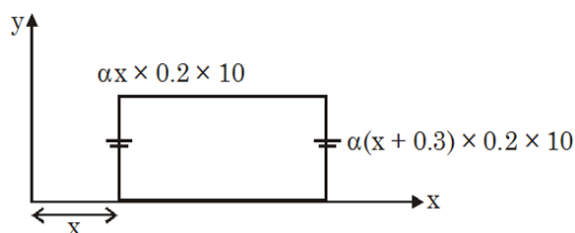
2. Ans (9.81 to 9.82)

$$Q = \frac{\omega m}{b} = \frac{10\pi}{3.2} = 9.817$$

$$\frac{b}{2m} = r = 1.6$$

$$\frac{b}{m} = 3.2$$

3. Ans (0.15)



$$\varepsilon_{\text{net}} = 0.6 \alpha = 0.12$$

$$i = \frac{0.12}{0.8} = \frac{12}{80} = 0.15 \text{ A}$$

4. Ans (18.00)

$$P = i^2 R = 0.15^2 \times 0.8 = 225 \times 8 \times 10^{-5}$$

$$= 1.8 \times 10^{-2} \text{ J} = 18 \text{ mJ}$$

5. Ans (-10.00)

$$\frac{1}{2} Li^2 = \frac{1}{2} (L_0 + L_k) i^2$$

$$\frac{1}{2} L_k i^2 = \frac{1}{2} m_e \times n \times A \ell v^2$$

$$i = neAv$$

$$\frac{1}{2} L_k n^2 e^2 A^2 v^2 = \frac{1}{2} m_e n A \ell v^2$$

$$L_k = \frac{m_e \ell'}{ne^2 A} = \frac{m_e 2\pi R \times n_e \ell}{ne^2 A}$$

$$L = \mu_0 n_e^2 A_c \ell$$

$$\frac{L_k}{L} = \frac{m_e \times 2\pi R_L n_e \ell}{ne^2 A} \times \frac{1}{\mu_0 n_e^2 A_c \ell}$$

$$= \frac{9.1 \times 10^{-31} \times 2\pi \times 10^{-2}}{10^{28} \times (1.6 \times 10^{-19})^2 \times 10^{-6} \times 4\pi \times 10^{-7} \times 10^3 \times \pi \times 10^{-4}}$$

$$= \frac{9.1}{2\pi \times 256} \times 10^{-9} \approx 10^{-10}$$

6. Ans (4.40 to 4.50)

$$\varepsilon = L \frac{di}{dt} = \frac{n_e \ell}{ne^2 A} \times i_0 \omega \cos \omega t$$

$$1 = \frac{9.1 \times 10^{-31} \times 2\pi \times 10^{-2} \times 1000 \times 10}{10^{28} \times (1.6 \times 10^{-19})^2 \times 10^{-6}} \times 1000 \times \omega$$

$$1 = \frac{9.1 \times 2\pi \times 10^{-26}}{2.56 \times 10^{-16}} \omega$$

$$\omega = \frac{2.56}{18.2\pi} \times 10^{10} = 4.47 \times 10^8$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (100)

$$\tan \phi = \frac{\omega L - \frac{1}{\omega C}}{R} = \frac{\omega^2 LC - 1}{\omega RC}$$

$$\omega = \sqrt{\omega_0^2 - 2r^2} = \sqrt{\frac{1}{LC} - \frac{R^2}{2L^2}}$$

$$\tan \phi = \frac{\omega^2 LC - 1}{\omega RC} = -\frac{\frac{R^2 C}{2L}}{\omega RC} = -\frac{R}{2L\omega}$$

$$\frac{R}{2L} = r; \tan \phi = -\frac{r}{\omega}$$

$$\tan \phi = -\frac{0.1}{10} = 0.1 = 0.01$$

$$\cot \phi = 100$$

8. Ans (3600)

Magnetic field on the axis of a circular loop

$$B = \left(\frac{\mu_0}{4\pi} \right) \times \frac{2\pi I R^2}{(R^2 + z^2)^{3/2}}$$

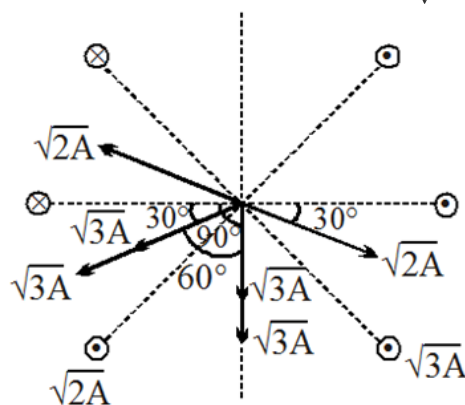
where R = radius of loop = $3 \times 10^{-2} \text{ m}$

$$10^{-7} \times \frac{2\pi \times 2.5 \times 3^2 \times 10^{-4}}{125 \times 10^{-6}} \hat{k}$$

$$= \left(\frac{9\pi}{25} \times 10^{-5} \text{ T} \right) \hat{k} = (36\pi \times 10^{-7} \text{ T}) \hat{k}$$

9. Ans (640)

$$B_1 = \frac{\mu_0}{4\pi} \times \frac{\sqrt{3} \times 2}{3 \times 10^{-2}} = 10^{-5} \times \frac{2}{\sqrt{3}} = \frac{2}{\sqrt{3}} \times 10^{-5}$$



$$B = \sqrt{(2B_1)^2 + (2B_2)^2 + 2(2B_1) \times (2B_2) \cos 60^\circ}$$

$$= B \sqrt{4 + 4 + 4} = 2\sqrt{3} \times \frac{2}{\sqrt{3}} \times 10^{-5}$$

$$= 4 \times 10^{-5} = 40 \times 10^{-6} = 640$$

PART-2 : CHEMISTRY

SECTION-I (i)

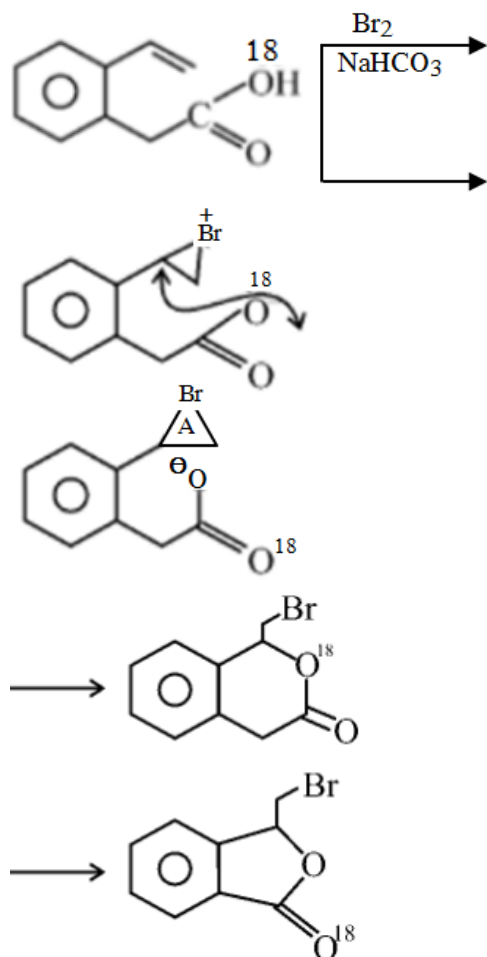
2. Ans (A,B,D)

$$\Delta S + \Delta S_{\text{surr}} \geq [= \text{Rev}, > \text{Irr}]$$

$$(\Delta S)_{U,V} \geq 0 [= \text{Rev}, > \text{Irr}]$$

5. Ans (A,B)

(A,B)



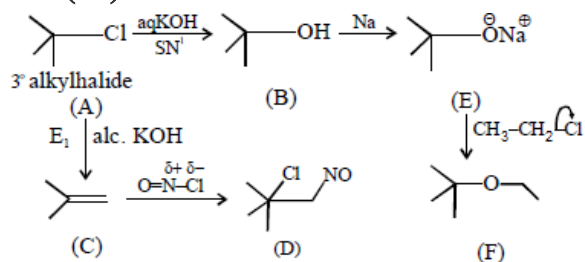
6. Ans (A,B,D)

:CF₂ Abnormal Reimer Tiemann reaction

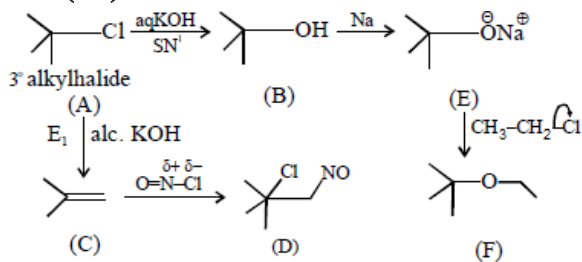
PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (B)



10. Ans (D)



PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (2.30)

$$\text{pH at I half equivalence point} = \text{p}K_{a_3} = 12$$

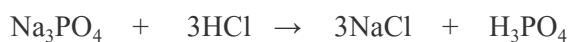
$$\text{pH at I equivalence point} = \frac{\text{p}K_{a_2} + \text{p}K_{a_3}}{2} = 10$$

$$\Rightarrow \text{p}K_{a_2} = 8$$

$$\text{pH at II equivalence point} = \frac{\text{p}K_{a_1} + \text{p}K_{a_2}}{2} = 6$$

$$\Rightarrow \text{p}K_{a_1} = 4$$

at III equivalence point



$$\begin{array}{cccc} 100 & 300 & - & - \\ \times & \times & 100 & 100 \end{array}$$

$$C = \frac{100}{400} = \frac{1}{4} \text{M}$$

$$K_{a_1} = 10^{-4} = \frac{(\text{H}^+)^2}{C} \Rightarrow (\text{H}^+) = \sqrt{K_{a_1} C}$$

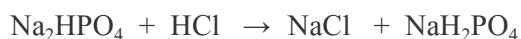
$$= \frac{10^{-2}}{2}$$

$$= 5 \times 10^{-3}$$

$$\text{pH} = 3 - \log 5$$

2. Ans (8.00)

at II half equivalence point



$$\begin{array}{cccc} 100 & 50 & - & - \\ 50 & \times & 50 & 50 \end{array}$$

$$\text{pH} = \text{p}K_{a_2} + \log \frac{50}{50} = \text{p}K_{a_2} = 8$$

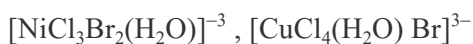
PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (13)

$$\begin{aligned}(\Delta x)_{\min} &= \frac{h}{(4\pi m \Delta V)} \\&= \frac{h}{4\pi m \left(\sqrt{\frac{2KE}{m}} \times \frac{1}{100} \right)} \\&= \frac{h \times 100}{4\pi \sqrt{2mKE}} \\&= \frac{6.6 \times 10^{-34} \times 100}{4 \times \pi \sqrt{2 \times (4 \times 1.66 \times 10^{-27})(0.4 \times 10^6 \times 1.6 \times 10^{-19})}} \\&= \sqrt{\frac{6.6 \times 10^{34+2}}{4 \times 3.14 \times 2.91 \times 10^{-20}}} \\&= 1.80 \times 10^{-13} \text{ m.}\end{aligned}$$

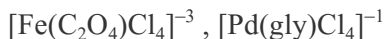
9. Ans (2)



show only geometrical isomerism



optical isomerism.



Show neither optical nor geometrical.

PART-3 : MATHEMATICS

SECTION-I (i)

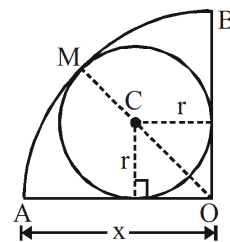
1. Ans (C)

$$\begin{aligned}(\theta_1 + \theta_2) \left(\frac{\cos^4 x}{\theta_1} + \frac{\sin^4 x}{\theta_2} \right) &= 1 \\ \cos^4 x + \sin^4 x + \frac{\theta_2}{\theta_1} \cos^4 x + \frac{\theta_1}{\theta_2} \sin^4 x &= 1 \\ 1 - 2\sin^2 x \cos^2 x + \frac{\theta_2}{\theta_1} \cos^4 x + \frac{\theta_1}{\theta_2} \sin^4 x &= 1 \\ \left(\sqrt{\frac{\theta_2}{\theta_1}} \cos^2 x - \sqrt{\frac{\theta_1}{\theta_2}} \sin^2 x \right)^2 &= 0 \\ \tan^2 x &= \frac{\theta_2}{\theta_1} \\ \theta_2 &= \theta_1 \tan^2 x \\ \frac{d\theta_2}{d\theta_1} &= \tan^2 x\end{aligned}$$

2. Ans (A,C,D)

$$\text{OM} = \text{OC} + \text{CM}$$

$$\Rightarrow x = r\sqrt{2} + r$$



$$\Rightarrow \frac{x}{r} = \sqrt{2} + 1 \Rightarrow \frac{r}{x} = \sqrt{2} - 1$$

$$\Rightarrow \frac{dx}{dr} = \sqrt{2} + 1 \text{ and } \frac{dr}{dx} = \sqrt{2} - 1$$

$$\text{and } \frac{d^2x}{dr^2} = 0$$

Now

$$\left(\frac{dr}{dx} \right)^2 + \left(\frac{dx}{dr} \right)^2 = (\sqrt{2} + 1)^2 + (\sqrt{2} - 1)^2 = 2(2 + 1) = 6$$

$$\& \frac{x+r}{x-r} = \frac{\sqrt{2}+2}{\sqrt{2}} \text{ and } \frac{r-x}{r+x} = \frac{\sqrt{2}-2}{\sqrt{2}}$$

$$\Rightarrow \frac{x+r}{x-r} + \frac{r-x}{r+x} = 2$$

3. Ans (A,C,D)

$$f(|x|) = |x| - 2, 0 \leq |x| \leq 3 \Rightarrow f(|x|)$$

$$= \begin{cases} -x - 2, & -3 \leq x \leq 0 \\ x - 2, & 0 < x \leq 3 \end{cases}$$

$$\Rightarrow [f(|x|)] = \begin{cases} [-x] - 2, & -3 \leq x \leq 0 \\ [x] - 2, & 0 < x \leq 3 \end{cases}$$

Also

$$|f(x)| = \begin{cases} 2, & -3 \leq x \leq 0 \\ 2 - x, & 0 < x \leq 2 \\ x - 2, & 2 < x \leq 3 \end{cases}$$

$$\Rightarrow [f(x)] = \begin{cases} 2, & -3 \leq x \leq 0 \\ 2 + [-x], & 0 < x \leq 2 \\ [x] - 2, & 2 < x \leq 3 \end{cases}$$

$$\text{So } [f(x)] + [f(x)] = \begin{cases} [-x] & -3 \leq x \leq 0 \\ [x] + [-x], & 0 < x \leq 2 \\ 2[x] - 4, & 2 < x \leq 3 \end{cases}$$

Now, options can be checked directly without further simplification.

4. Ans (A,B,C,D)

$$y \sin \theta = x$$

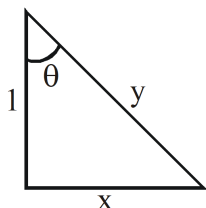
$$y \cos \theta = 1$$

$$(y^n - x^n) = \frac{1 - \sin^n \theta}{\cos^n \theta}$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin \theta}{\cos \theta} = 0,$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin^2 \theta}{\cos^2 \theta} = 1$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin^3 \theta}{\cos^3 \theta} = \infty \Rightarrow \text{Limit does not exist.}$$



5. Ans (A,C)

Differentiating $fg = 1$, we get $f'g + g'f = 0 \Rightarrow$

$$\frac{f'}{f} = -\frac{g'}{g}$$

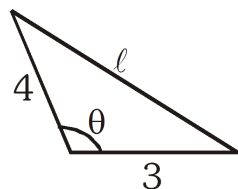
Differentiating again, we get $f''g + 2f'g' + g''f = 0$

$$f'f = 0 \Rightarrow \frac{f''}{f'} \cdot \frac{g}{g'} + 2 + \frac{g''}{g'} \cdot \frac{f}{f'} = 0$$

$$\Rightarrow \frac{f''}{f'} \cdot \frac{g}{g'} + 2 - \frac{g''}{g'} \cdot \frac{g}{g'} = 0 \Rightarrow \frac{g}{g'} \left(\frac{f''}{f'} - \frac{g''}{g'} \right)$$

$$= -2 \Rightarrow \left(\frac{f''}{f'} - \frac{g''}{g'} \right) = \frac{-2g'}{g} \text{ or } \frac{2f'}{f}$$

6. Ans (B)



Let angle between the hands be θ and distance between tips be ℓ

$$\ell = \sqrt{25 - 24 \cos \theta}$$

we have to find ℓ when $\frac{d\ell}{d\theta}$ is maximum

$$\frac{d\ell}{d\theta} = \frac{12 \sin \theta}{\sqrt{25 - 24 \cos \theta}} \text{ to maximum } \frac{d\ell}{d\theta}$$

$$\frac{d^2\ell}{d\theta^2} = 12 \left\{ \frac{\sqrt{25 - 24 \cos \theta} \cdot \cos \theta - \frac{24 \sin^2 \theta}{2\sqrt{25 - 24 \cos \theta}}}{(25 - 24 \cos \theta)} \right\} = 0$$

$$(25 - 24 \cos \theta) \cos \theta - 12 \sin^2 \theta = 0$$

$$12 \cos^2 \theta - 25 \cos \theta + 12 = 0$$

$$\cos \theta = \frac{3}{4} \quad \cos \theta = \frac{4}{3} \quad (\text{Not possible})$$

$$\text{hence } \ell = \sqrt{25 - 24 \cdot \frac{3}{4}} = \sqrt{7}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (D)

$$f(x) = [P(x)] + [P(x)]$$

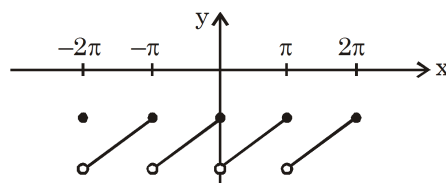
$$\Rightarrow f(x) = \begin{cases} 0 & x = n\pi \\ -1 & x \neq n\pi \end{cases}$$

$$g(x) = \left\{ \frac{x}{\pi} \right\} - 1$$

$$\Rightarrow f(x) = g(x) \text{ has no solution, } \lim_{x \rightarrow 0} f(x) = -1$$

$f(x)$, $g(x)$ & $h(x)$ are periodic with period π ,

graph of $h(x) = f(x) + g(x)$



so $h(x)$ is not differentiable

at 4 points in $x \in [-2\pi, 2\pi]$

8. Ans (C)

$$f(x) = [P(x)] + [P(x)]$$

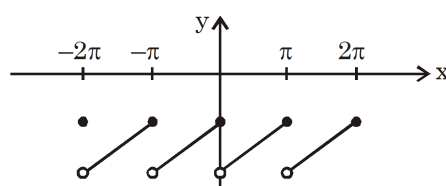
$$\Rightarrow f(x) = \begin{cases} 0 & x = n\pi \\ -1 & x \neq n\pi \end{cases}$$

$$g(x) = \left\{ \frac{x}{\pi} \right\} - 1$$

$$\Rightarrow f(x) = g(x) \text{ has no solution, } \lim_{x \rightarrow 0} f(x) = -1$$

$f(x)$, $g(x)$ & $h(x)$ are periodic with period π ,

graph of $h(x) = f(x) + g(x)$



so $h(x)$ is not differentiable

at 4 points in $x \in [-2\pi, 2\pi]$

9. Ans (B)

$$\frac{a+b+c}{b} = 2 \left(\frac{f(1)}{f(1) - f(-1)} \right)$$

$$= 2 \left(\frac{1}{1 - \frac{f(-1)}{f(1)}} \right)$$

$$\text{minimum of } \frac{a+b+c}{b} = 2 \text{ as } \frac{f(-1)}{f(1)} \geq 0$$

10. Ans (B)

$$H(x) = (x - \alpha)(x - 1)(x - 2)(x - 3) + x$$

$$\Rightarrow H(-1) = (-1 - \alpha)(-2)(-3)(-4) - 1 = 24(1 + \alpha) - 1$$

$$H(5) = (5 - \alpha)4 \times 3 \times 2 + 5 = 24(5 - \alpha) + 5$$

$$H(-1) + H(5) = 148$$

$$\text{Similarly } H(0) + H(4) = 28$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (2.50)

$$h(x) = \begin{cases} f(x) & x^2 < 1 \\ \frac{f(x)+g(x)}{2} & x^2 = 1 \\ g(x) & x^2 > 1 \end{cases}$$

2. Ans (0.93 or 0.94)

$$h(x) = \begin{cases} f(x) & x^2 < 1 \\ \frac{f(x)+g(x)}{2} & x^2 = 1 \\ g(x) & x^2 > 1 \end{cases}$$

3. Ans (2.00)

$$\left[\frac{x+1}{x} > 0 \right] \Rightarrow \left[x \in (-\infty, -1) \cup (0, \infty) \right] \dots (1)$$

$$\left[\left| \ln \left(\frac{x+1}{x} \right) \right| \leq 1 \right] \Rightarrow \left[\frac{1}{e} \leq \frac{x+1}{x} \leq e \right] \dots (2)$$

$$\text{From (2), } \frac{x+1-ex}{x} \leq 0 \text{ and } \frac{ex+e-x}{ex} \geq 0$$

$$\Rightarrow \frac{x(e-1)-1}{x} \geq 0 \text{ and } \frac{x(e-1)+e}{ex} \geq 0$$

$$x \in (-\infty, 0) \cup \left[\frac{1}{e-1}, \infty \right)$$

$$\text{and } x \in \left(-\infty, \frac{e}{1-e} \right] \cup (0, \infty)$$

from (1) and (2)

$$x \in \left(-\infty, \frac{e}{1-e} \right] \cup \left[\frac{1}{e-1}, \infty \right)$$

and range of $f(x)$ is $[-1, 1]$

4. Ans (3.00)

$$\left[\frac{x+1}{x} > 0 \right] \Rightarrow \left[x \in (-\infty, -1) \cup (0, \infty) \right] \dots (1)$$

$$\left[\left| \ln \left(\frac{x+1}{x} \right) \right| \leq 1 \right] \Rightarrow \left[\frac{1}{e} \leq \frac{x+1}{x} \leq e \right] \dots (2)$$

$$\text{From (2), } \frac{x+1-ex}{x} \leq 0 \text{ and } \frac{ex+e-x}{ex} \geq 0$$

$$\Rightarrow \frac{x(e-1)-1}{x} \geq 0 \text{ and } \frac{x(e-1)+e}{ex} \geq 0$$

$$x \in (-\infty, 0) \cup \left[\frac{1}{e-1}, \infty \right)$$

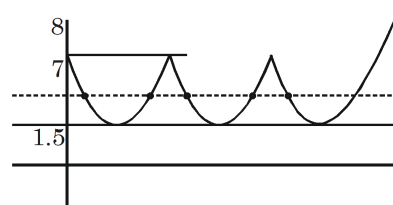
$$\text{and } x \in \left(-\infty, \frac{e}{1-e} \right] \cup (0, \infty)$$

from (1) and (2)

$$x \in \left(-\infty, \frac{e}{1-e} \right] \cup \left[\frac{1}{e-1}, \infty \right)$$

and range of $f(x)$ is $[-1, 1]$

5. Ans (6.00)



$$\text{min. value of } g(1) = \frac{-11}{2}$$

$f_1(x)$ has 2 non-differentiable points

$f_2(x)$ has 4 non-differentiable points

$$f_2(x) = ||g(x)| - 7| = ||6x^2 - 18x + 8| - 7| \text{ for } f_3(x)$$

to have 10-nondifferentiable points

$$P_2 \in (1.5, 7)$$

$$5 < P_1 < P_2 < P_3 < P_4 < \dots$$

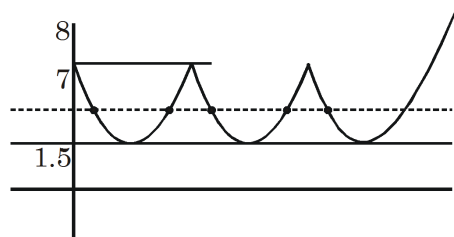
$$\Rightarrow a_1 = 2, a_2 = 4, a_3 = 6$$

$$H(x) = \frac{(x-2)(x-6)}{(x-4)} = y$$

$$\Rightarrow x^2 - x(8+y) + 4y + 12 = 0$$

$$Q(x) \geq 0 \Rightarrow y^2 + 16 \geq 0 \Rightarrow y \in \mathbb{R}$$

6. Ans (2.00)



$$\text{min. value of } g(1) = \frac{-11}{2}$$

$f_1(x)$ has 2 non-differentiable points

$f_2(x)$ has 4 non-differentiable points

$$f_2(x) = ||g(x)| - 7| = ||6x^2 - 18x + 8| - 7| \text{ for } f_3(x)$$

to have 10-nondifferentiable points

$$P_2 \in (1.5, 7)$$

$$5 < P_1 < P_2 < P_3 < P_4 < \dots$$

$$\Rightarrow a_1 = 2, a_2 = 4, a_3 = 6$$

$$H(x) = \frac{(x-2)(x-6)}{(x-4)} = y$$

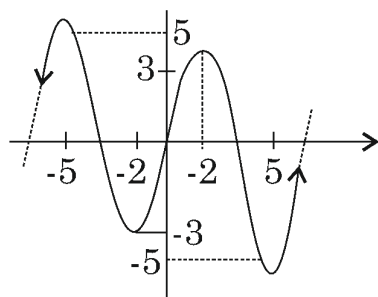
$$\Rightarrow x^2 - x(8+y) + 4y + 12 = 0$$

$$Q(x) \geq 0 \Rightarrow y^2 + 16 \geq 0 \Rightarrow y \in \mathbb{R}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (5)



$$f(x) = ax^9 + b \sin x + cx^2 \operatorname{sgn}(x) + \frac{e^x - e^{-x}}{e^x + e^{-x}}; a > 0 \text{ \& } b, c \in \mathbb{R}$$

is an odd function (check yourself)

$$\Rightarrow f(0) = 0$$

$$f(-5) = 5, f(-2) = -3 \Rightarrow f(5) = -5, f(2) = 3$$

Also, as $x \rightarrow \infty$; $f(x) \rightarrow \infty$ and $x \rightarrow -\infty$; $f(x) \rightarrow -\infty$

Hence no of zeroes of $f(x) = 0$ is 5

8. Ans (1)

$$f'(0^+) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

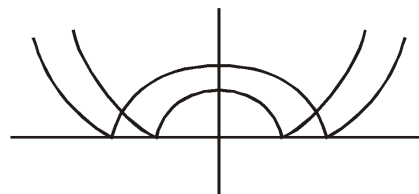
$$= \lim_{h \rightarrow 0} \frac{\frac{\sin^{-1}(h^2+h^3)-h}{h} + 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(h^2+h^3)}{h^2} \cdot \frac{(h^2+h^3)}{(h^2+h^3)} = 1$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} \frac{\sin^{-1}(h^2+h^3) - h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{\sqrt{1-(h^2+h^3)^2}} \cdot (2h+3h^2) - 1}{1} = -1$$

9. Ans (32)



Point of intersection when

$$x^2 - 1 = 3 - x^2 \Rightarrow x = \sqrt{2}$$

$$y = x^2 - 1 \text{ \& } y = 3 - x^2$$

$$\left(\frac{dy}{dx}\right)_1 = 2\sqrt{2} \text{ \& } \left(\frac{dy}{dx}\right)_2 = -2\sqrt{2}$$

$$\tan \theta = \frac{4\sqrt{2}}{1-8} \Rightarrow \theta = \tan^{-1} \left(\frac{4\sqrt{2}}{7} \right)$$